

Veille informatique quantique - 2026-09-21

Période analysée : les 7 derniers jours. Articles retenus : 10.

Synthèse

1. Performance of the spin qubit shuttling architecture for a surface code implementation

Source : [arXiv Quantum Physics](#) - publié le 2026-09-21

Scores : pertinence 87/100 | nouveauté 50/100 | importance 50/100 | score 67/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2503.10601v3 Announce Type: replace Abstract: Qubit shuttling promises to advance some quantum computing platforms to the qubit register sizes needed for effective quantum error correction (QEC), but also introduces additional errors whose impact must be evaluated. The established method to investigate the performance of QEC codes in a realistic scenario is to employ a standard noise model known as circuit-level noise, where all quantum operations are modeled as noisy. In the present work

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum computing, qubit, quantum error correction, quantum processor.

2. Shock-Capturing Quantum Algorithm for the Linear Advection Operator

Source : [arXiv Quantum Physics](#) - publié le 2026-09-21

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2609.21153v1 Announce Type: new Abstract: The linear advection operator is an ubiquitous building block in fluid and plasma problems. We develop a quantum algorithm for enacting the operator. When the advection velocity is constant in space, our algorithm is exponentially more efficient per time step than classical and avoids spurious oscillations near steep gradients. The algorithm is most cleanly illustrated using the one-dimensional advection equation on a uniform spatial grid with per

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Mots-clés détectés : qubit, quantum algorithm, Quantinuum.

3. Readout-induced degradation of transmon lifetimes: interplay of TLSs and qubit spectral reshaping

Source : [arXiv Quantum Physics](#) - publié le 2026-09-21

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2603.13857v2 Announce Type: replace Abstract: Measurement backaction degrades dispersive readout of superconducting qubits even at modest drive strengths, often via the reduction of qubit lifetimes during readout. In this work, we theoretically and experimentally study this degradation and show how it can result from the interplay between detuned two-level systems (TLSs) and a drive-renormalized qubit spectrum. For modest to strong readout, the qubit emission spectrum becomes non-Lorentzi

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Mots-clés détectés : qubit, superconducting qubit, quantum processor.

4. Grover Search with Semiconductor Spin Qubits at Ambient Conditions

Source : [arXiv Quantum Physics](#) - publié le 2026-09-21

Scores : pertinence 79/100 | nouveauté 50/100 | importance 50/100 | score 63/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2609.13022v2 Announce Type: replace Abstract: Grover's algorithm is executed on a commercial quantum computer based on nitrogen-vacancy centers in diamond operating under ambient conditions, achieving fidelities up to 99.98% . Within a $N=8$ search space of three solid-state nuclear spin qubits, the measured success probabilities of finding a single or two marked states are $(77.3 \pm 3.4)\%$ and $(87.0 \pm 4.2)\%$, respectively. These values surpass published results for supercon

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Mots-clés détectés : quantum computer, qubit, superconducting qubit.

5. Oracle Synthesis Based on X-Map Decision Diagrams

Source : [arXiv Quantum Physics](#) - publié le 2026-09-21

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2609.20854v1 Announce Type: new Abstract: Quantum oracles act as reversible black-box operators that encode classical Boolean functions into quantum states, enabling efficient function evaluation in quantum superposition. The resource efficiency of oracle implementation is critical to the performance of numerous quantum algorithms. Most state-of-the-art oracle synthesis approaches rely on compact Boolean function representations such as exclusive-sum-of-products (ESOP), yet still suffer f

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Mots-clés détectés : quantum computing, quantum algorithm.

6. Hybrid quantum-classical attention for histopathology-based molecular profiling in data-limited cancers

Source : [arXiv Quantum Physics](#) - publié le 2026-09-21

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2609.21115v1 Announce Type: new Abstract: Molecular profiling from routine histopathology could expand access to precision oncology when sequencing is unavailable, tissue is limited, or training cohorts are small. We developed a hybrid quantum-classical strategy that replaces softmax attention in a transformer for histopathology-based gene expression prediction with a quantum-derived doubly stochastic matrix (QDSM). Across 29 cancer cohorts from The Cancer Genome Atlas and an independent

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : quantum processor, IBM Quantum.

7. The Pinnacle Architecture with fixed connectivity of degree eight

Source : [arXiv Quantum Physics](#) - publié le 2026-09-21

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2609.21249v1 Announce Type: new Abstract: We show how the Pinnacle architecture can be adapted to be compatible with superconducting qubits, where the connectivity between qubits must be fixed at fabrication. Specifically, we present a modified instantiation of the architecture that has fixed connectivity degree of eight. With this instantiation, we show that a 2048-bit RSA integer can be factored in one month with approximately 120 000 physical qubits, given a physical error rate of 10^{-4}

Pourquoi c'est important : Article sélectionné par les mots-clés de la veille.

Mots-clés détectés : qubit, superconducting qubit.

8. Time rescaling for second-order feedback-based quantum optimization

Source : [arXiv Quantum Physics](#) - publié le 2026-09-21

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2609.21640v1 Announce Type: new Abstract: Feedback-based quantum algorithms recently gained attention by demonstrating how quantum computers can be used to solve optimization problems without resorting to hybrid quantum-classical architectures. Recent progress made with Feedback-based Algorithm for Quantum Optimization (FALQON) variants significantly reduced the depth of the required circuits. In this work, we merge the time-rescaled and second-order FALQON variants into a unified framework

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Mots-clés détectés : quantum computer, quantum algorithm.

9. Single-atom-based asynchronous photonic interconnect for scalable modular quantum computing

Source : [arXiv Quantum Physics](#) - publié le 2026-09-21

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2609.21961v1 Announce Type: new Abstract: Scaling quantum computation beyond the capacity of a single quantum processing unit requires quantum interconnects between modular processors. Optical photons are natural carriers for distributing entanglement between these processors. Most loss-resilient protocols use photonic Bell-state measurements based on the linear-optics type-II fusion gate. The resulting entanglement rate scales quadratically with each processor's typically low photon-delivery

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Mots-clés détectés : quantum computing, qubit.

10. Enumeration of all superconducting circuits up to 5 nodes

Source : [arXiv Quantum Physics](#) - publié le 2026-09-21

Scores : pertinence 71/100 | nouveauté 50/100 | importance 50/100 | score 59/100.

Catégorie : recherche - Qualité de la source : secondaire.

arXiv:2410.18497v2 Announce Type: replace Abstract: Nonlinear superconducting circuits can be used as amplifiers, transducers, and qubits. Only a handful have been analyzed or built, so useful configurations likely remain undiscovered. In this paper, we enumerate all unique nonlinear superconducting circuits, up to five nodes in size, built of capacitors, inductors, and Josephson junctions. We catalog the resulting design space by sorting circuits into "Hamiltonian classes" using a set of refer

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Mots-clés détectés : qubit, superconducting qubit.

Méthode

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